

## Grain to Glass Without the Grain: Developing a Midwestern Rice Spirit with Traditional Koji Saccharification

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### Abstract

The American craft spirits market has shown growing interest in rice-based distillates occupying the conceptual space between East Asian soju and Western grain whiskey. This presentation describes the development of a production-scale rice spirit at Heartland Spirits LLC, with specific attention to three technical challenges: sourcing commercially viable Midwestern short-grain rice, adapting traditional *Aspergillus oryzae* (koji) saccharification protocols for a small-batch American distillery context, and navigating TTB formula approval for a product that does not fit cleanly into existing Class/Type designations.

Beginning in November 2025, pilot-scale batches (R&D batch HB-2026-018) were developed in collaboration with Dr. Patricia Liang, UW-Madison Department of Food Science. Koji cultivation employed a modified cedar-box protocol at 86F with humidity control adapted from sake production literature. Glucose and saccharification rate data from five cultivation runs are presented, demonstrating the technical feasibility of koji saccharification at 5-gallon pilot scale with grain sourced from a single Midwest supplier.

We further discuss the TTB classification question: whether a koji-saccharified rice distillate aged briefly in used American oak qualifies as 'whisky' under 27 CFR 5.22(b)(1) or must be labeled as a 'distilled spirits specialty.' The regulatory path chosen -- and its implications for labeling, pricing, and market positioning -- will be reviewed in the context of other recent producer filings.

The presentation concludes with sensory and gas chromatography data on the pilot distillate and a discussion of scale-up considerations for a 2027 commercial release.

### Keywords

rice spirit; koji; *Aspergillus oryzae*; TTB classification; craft distilling; Midwestern grain

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